# Wei-Kai Li Academic CV

Professor / Doctoral Supervisor / Taishan Scholars Young Expert

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School of Computer Science and Artificial Intelligence, Shandong Jianzhu University / School of Mathematics and Statistics, Chongqing Jiaotong University / Panoramic Medical Imaging Center



## Research Interests

- Brain Networks & Neurological Disorders: Brain connectivity modeling and neurological disorder analysis.
- Transfer Learning & Domain Adaptation: Methods that transfer knowledge across domains and data sources.
- Medical Imaging & Computational Biology: Medical image analysis and drug-target interaction prediction.
- Computer Vision & Multimodal AI: Visual recognition, few-shot detection, and multimodal alignment.
- Intelligent Algorithms & Models: Multi-view clustering, neural networks, and intelligent learning methods.

## Education

1. 2018–2022 / PhD / Computer Science and Technology / Nanjing University of Aeronautics and Astronautics
2. 2015–2018 / Master's / Computer Technology and Applications / Chongqing Jiaotong University
3. 2011–2015 / Bachelor's / Information and Computing Science / Chongqing Jiaotong University

## Research Projects

1. Unsupervised Domain Adaptation beyond Domain Alignment and Its Extensions / NSFC Young Scientists Fund / Principal Investigator
2. Advanced School in Applied Machine Learning / NSFC International Cooperation and Exchange / Principal Investigator
3. Multimodal Neurophysiological Mechanisms of Heterogeneous Brain Networks in Depression and Digital-Twin Closed-Loop TMS Robotics / Shandong Major Basic Research Project / Principal Investigator
4. Exploration of Unsupervised Domain Adaptation beyond Domain Alignment / MIIT Key Laboratory Open Project / Principal Investigator
5. Early Identification of Parkinson's Disease Using PET/MR Multimodal Brain Networks / Hubei Key Laboratory of Molecular Imaging / Principal Investigator
6. Multiscale Analysis of Autism Using Multimodal Imaging / Chongqing Education Commission Young Research Fund / Principal Investigator
7. Brain Network Disease Diagnosis Based on Multicenter Transfer Learning / Chongqing Natural Science Foundation General Project / Principal Investigator
8. Probabilistic Synchronization and Secure Control of Stochastic Complex Networks / Chongqing Natural Science Foundation Joint Fund / Researcher
9. Investigating and Extending the Effectiveness of Self-Supervised Learning / NSFC General Program / Researcher

## Journal Articles

[1] WANG Q, XIA Y, WEI L, et al. Machine learning-based assessments of perioperative features in determining long-term conduct block post-transcatheter closure of ventricular septal defect[J]. Cardiology in the Young, 2026: 1-11.

[2] QIN H, HONG G, MA D, et al. Universum Inspired Graph Contrast Learning for Major Depressive Disorder Identification: In Functional Brain Network View[J]. CNS Neuroscience & Therapeutics, 2026, 32(9): e71158.

[3] YANG J, LI W, QIAN K, et al. Domain generalization meets mutual bi-attention: A novel approach to drug-target prediction[J]. Neurocomputing, 2026: 134626.

[4] CUI L, WU Y, TANG Y, et al. Enhancing Fatigue Detection Through Heterogeneous Multisource Data Integration and Cross-Domain Modality Imputation[J]. IEEE Transactions on Computational Social Systems, 2026.

[5] MA D, PENG L, ZHANG L, et al. Hierarchical neighbor integration graph attention network for autism spectrum disorder diagnosis[J]. Frontiers in Psychiatry, 2026, 17: 1846996.

[6] WANG H, LENG J, LI S, et al. TumorAL: Evidence-aware active learning for 3D tumor segmentation[J]. Neurocomputing, 2026: 134239.

[7] RUAN Y, GUAN L, PENG L, et al. Identification of Major Depressive Disorder Using Multiple Functional Connection Patterns[J]. CNS Neuroscience & Therapeutics, 2026, 32(6): e70951.

[8] WANG Q, LI W, WAN J, et al. Electrocardiographic Alterations Combined with Hematological, Biochemical, and Metabolic Profiles Predict Prognosis in Kawasaki Disease[J]. Journal of Cardiovascular Development and Disease, 2026, 13(6): 228.

[9] CHEN X, ZOU Q, CAI Q, et al. Partial domain adaptation enables cross domain cell type annotation between scRNA-seq and snRNA-seq[J]. PLOS Computational Biology, 2026, 22(5): e1014223.

[10] WU Y, LI Y, WEI H, et al. Test-time few-shot object detection via dynamic prototype fusion[J]. IEEE Transactions on Cybernetics, 2026.

[11] YANG Y, LI B, CAO W, et al. Cross-modal medical image generation from MRI to PET using robust generative adversarial network[J]. Expert Systems with Applications, 2026, 297: 129287.

[12] LI D, YAO J, LI W, et al. More insights into disruption and decoupling of individual metabolic connectomes in Parkinson's disease[J]. Progress in Neuro-Psychopharmacology and Biological Psychiatry, 2025: 111578.

[13] RUAN Y, MA H, MA D, et al. Low-light image enhancement using dual cross attention[J]. Engineering Applications of Artificial Intelligence, 2025, 159: 111501.

[14] MOU X, LI W, LI B, et al. Auto-weighted projective one-step multi-view clustering[J]. Neurocomputing, 2025: 131960.

[15] XU X, LI Y, LAN H, et al. Quantitative susceptibility mapping shows alterations of brain iron content in children with autism spectrum disorder: a whole-brain analysis[J]. BMC psychiatry, 2025, 25(1): 826.

[16] MA Q, CHEN B, SEIFERT R, et al. PSMA PET/CT based multimodal deep learning model for accurate prediction of pelvic lymph-node metastases in prostate cancer patients identified as candidates for extended pelvic lymph node dissection by preoperative nomograms[J]. European Journal of Nuclear Medicine and Molecular Imaging, 2025, 52(6): 2063-2075.

[17] CHEN X, LIN S, CHEN X, et al. Timestamp calibration for time-series single cell rna-seq expression data[J]. Journal of Molecular Biology, 2025, 437(9): 169021.

[18] CHEN X, LIN D, LI Z, et al. Iterative neural networks for improving memory capacity[J]. Neural Networks, 2025, 182: 106936.

[19] SUN D, LI W, YU H, et al. The hidden Markov model reveals the changes in brain dynamics among patients with end-stage renal disease under different dialysis methods[J]. Journal of Cyber-Physical-Social Intelligence, 2024, 3(1): 10-10.

[20] CHEN H, XU J, LI W, et al. The characteristic patterns of individual brain susceptibility networks underlie Alzheimer's disease and white matter hyperintensity-related cognitive impairment[J]. Translational Psychiatry, 2024, 14(1): 177.

[21] LI W, WEI H, WU Y, et al. TIDE: Test-time few-shot object detection[J]. IEEE Transactions on Systems, Man, and Cybernetics: Systems, 2024, 54(11): 6500-6509.

[22] ZHOU W, SUN M, XU X, et al. Multipattern graph convolutional network-based autism spectrum disorder identification[J]. Cerebral Cortex, 2024, 34(3): bhae064.

[23] YI T, LI W, WEI W, et al. Limbic/paralimbic connection weakening in preschool autism-spectrum disorder based on diffusion basis spectrum imaging[J]. The European journal of neuroscience, 2024.

[24] ZHOU R, SUN C, SUN M, et al. Altered intra-and inter-network connectivity in autism spectrum disorder[J]. Aging, 2024, 16.

[25] YU Y, LI W, WU J, et al. Machine learning models using symptoms and clinical variables to predict coronary artery disease on coronary angiography[J]. Advances in Interventional Cardiology/Postępy w Kardiologii Interwencyjnej, 2024.

[26] YANG J, XU X, SUN M, et al. Towards an accurate autism spectrum disorder diagnosis: multiple connectome views from fMRI data[J]. Cerebral Cortex, 2024, 34(1): bhad477.

[27] LI YL, WU JJ, LI WK, et al. Effects of individual metabolic brain network changes co-affected by T2DM and aging on the probabilities of T2DM: protective and risk factors[J]. Cerebral Cortex, 2024, 34(1): bhad439.

[28] XU X, CHEN P, LI W, et al. Topological properties analysis and identification of mild cognitive impairment based on individual morphological brain network connectome[J]. Cerebral Cortex, 2024, 34(1): bhad450.

[29] DING Y, XU X, PENG L, et al. Wavelet transform-based frequency self-adaptive model for functional brain network[J]. Cerebral Cortex, 2023, 33(22): 11181-11194.

[30] LI W, AL-AMIN MM, NAIR A. New insights into the disorder of brain connectivity in schizophrenia[J]. Frontiers in Neuroimaging, 2023, 2: 1266695.

[31] LI W, TANG Y, PENG L, et al. The reconfiguration pattern of individual brain metabolic connectome for Parkinson's disease identification[J]. MedComm, 2023, 4(4): e305.

[32] WANG Y, CHEN Q, LIU Y, et al. TIToK: A solution for bi-imbalanced unsupervised domain adaptation[J]. Neural Networks, 2023, 164: 81-90.

[33] LIU X, XU Z, LI W, et al. Clinical application of machine learning methods in psychiatric disorders[J]. Frontiers in psychiatry, 2023, 14: 1209615.

[34] LI W, WANG Z, HU S, et al. Functional and structural brain network construction, representation and application[J]. Frontiers in neuroscience, 2023, 17.

[35] LI W, CHEN S. Unsupervised domain adaptation with progressive adaptation of subspaces[J]. Pattern Recognition, 2022, 132: 108918.

[36] TANG Y, LI W, TAO L, et al. Machine learning-derived multimodal neuroimaging of presurgical target area to predict individual's seizure outcomes after epilepsy surgery[J]. Frontiers in Cell and Developmental Biology, 2022, 9: 669795.

[37] LI WK, CHEN YC, XU XW, et al. Human-Guided Functional Connectivity Network Estimation for Chronic Tinnitus Identification: A Modularity View[J]. IEEE journal of biomedical and health informatics, 2022.

[38] LI W, CHEN S. Partial Domain Adaptation without Domain Alignment[J]. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2022.

[39] CHEN H, LI W, SHENG X, et al. Machine learning based on the multimodal connectome can predict the preclinical stage of Alzheimer’s disease: a preliminary study[J]. European Radiology, 2022, 32(1): 448-459.

[40] LI W, XU X, WANG Z, et al. Multiple connection pattern combination from single-mode data for mild cognitive impairment identification[J]. Frontiers in Cell and Developmental Biology, 2021, 9: 782727.

[41] XU X, WANG T, LI W, et al. Morphological, structural, and functional networks highlight the role of the cortical-subcortical circuit in individuals with subjective cognitive decline[J]. Frontiers in Aging Neuroscience, 2021, 13: 688113.

[42] LIN X, LI WK, DONG G, et al. Characteristics of multimodal brain connectomics in patients with schizophrenia and the unaffected first-degree relatives[J]. Frontiers in cell and developmental biology, 2021, 9: 631864.

[43] XU X, LI W, TAO M, et al. Effective and accurate diagnosis of subjective cognitive decline based on functional connection and graph theory view[J]. Frontiers in neuroscience, 2020, 14: 577887.

[44] LI W, XU X, JIANG W, et al. Functional connectivity network estimation with an inter-similarity prior for mild cognitive impairment classification[J]. Aging (Albany NY), 2020, 12(17): 17328.

[45] ZHANG F, LI W, LI H, et al. The effect of jet lag on the human brain: A neuroimaging study[J]. Human brain mapping, 2020, 41(9): 2281-2291.

[46] GAO X, XU X, HUA X, et al. Group similarity constraint functional brain network estimation for mild cognitive impairment classification[J]. Frontiers in Neuroscience, 2020, 14: 165.

[47] XU X, LI W, MEI J, et al. Feature selection and combination of information in the functional brain connectome for discrimination of mild cognitive impairment and analyses of altered brain patterns[J]. Frontiers in aging neuroscience, 2020, 12: 28.

[48] LIN DONGYUAN, CHEN XIAOFENG, SUN WENTAO, et al. Analysis of stability of time-delay exchangeable quaternion neural networks[J]. Journal of Smart Science and Technology, 2020, 2(1): 80-87.

[49] LI W, ZHANG L, QIAO L, et al. Toward a better estimation of functional brain network for mild cognitive impairment identification: a transfer learning view[J]. IEEE journal of biomedical and health informatics, 2019, 24(4): 1160-1168.

[50] LIU R, CHEN H, QIN R, et al. The altered reconfiguration pattern of brain modular architecture regulates cognitive function in cerebral small vessel disease[J]. Frontiers in neurology, 2019, 10: 324.

[51] LI W, QIAO L, ZHANG L, et al. Functional brain network estimation with time series self-scrubbing[J]. IEEE journal of biomedical and health informatics, 2019, 23(6): 2494-2504.

[52] LI W, GAO X, JI TONGJIAN, et al. A group-similarity-constrained functional brain network modeling method for mild cognitive impairment diagnosis[J]. Journal of Smart Science and Technology, 2019(2).

[53] LI W, WANG Z, JIANG W. An adaptive fractional-order partial differential image enhancement model[J]. Computer Engineering and Science, 2018, 40(4): 699.

[54] ZHOU Y, QIAO L, LI W, et al. Simultaneous estimation of low-and high-order functional connectivity for identifying mild cognitive impairment[J]. Frontiers in neuroinformatics, 2018, 12: 337293.

[55] LI W, WANG Z, ZHANG L, et al. Remodeling Pearson's correlation for functional brain network estimation and autism spectrum disorder identification[J]. Frontiers in neuroinformatics, 2017, 11: 55.

## Conference Papers

[56] CUI J, YIN J, LI W, et al. Anatomy-Preserving Schrodinger Bridge for Unpaired MRI Quality Enhancement[C]//2026 International Conference on Cyber-Physical Social Intelligence (CPSI). 2026: 1-6.

[57] CUI L, WU Y, XIAO W, et al. Fixed-Target Reuse of Sensor-Disjoint Human-Sensing Data via Bridge-Guided Signal-Format Conversion[C]//2026 International Conference on Cyber-Physical Social Intelligence (CPSI). 2026: 1-6.

[58] LEI X, ZHU J, LI W, et al. A Speckle Mutual Information-Based Network for Imaging Through Dynamic Scattering Media[C]//2026 International Conference on Cyber-Physical Social Intelligence (CPSI). 2026: 1-7.

[59] YANG J, ZHANG J, QIAN K, et al. Bridging the modality reliability gap in drug-target interaction prediction via a confidence-aware multimodal fusion framework[J]. Proceedings of the AAAI Conference on Artificial Intelligence, 2026, 40(32): 27529.

[60] MA H, GAN HS, MI H, et al. Attention Transfer Based Hybrid Knowledge Distillation for Multimodal Brain Tumor Segmentation[C]//2025 IEEE International Conference on Big Data (BigData). 2025: 7503-7510.

[61] SONG H, QIN H, LI W. Incorporating Negative Node Information in Graph Transformer for MDD Diagnosis[C]//2025 International Conference on Cyber-Physical Social Intelligence (CPSI). 2025: 1-6.

[62] LU Y, TIAN N, LI W. MVSSL-GT: A Graph Transformer Method with Multi-View Contrastive Self-Supervised Learning[C]//2025 International Conference on Cyber-Physical Social Intelligence (CPSI). 2025: 1-6.

[63] LI W, TIAN N, LI Y, et al. Revisiting Multi-Modal Alignment: In Distribution View[C]//2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC). 2025: 6325-6330.

[64] CHEN S, CHEN X, LI W. An Improved Algorithm for Spiking Neural Networks with Multi-Scale Attention Coding[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.

[65] XIAO X, CHEN X, LI W. Improved domain-adversarial neural networks with mixup and pseudo-labeling[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.

[66] JIANG Y, LI W, CHEN X. Dynamic Graph Contrastive Learning for Neuro Disease Identification using Resting-State fMRI[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.

[67] YANG Y, LI B, LI W, et al. Synthesizing Aβ-PET based on multi-modal neuroimaging fusion for pathological diagnosis of Alzheimer’s disease[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.

[68] QIN H, XU X, LI W. Universum Inspired Graph Contrast Learning for Autism Spectrum Disorder Identification[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.

[69] SUN D, CHEN X, LI W. Enhancing Functional Brain Network Construction for Neurological Disorder Diagnosis through Modularity-Driven Regularization[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.

[70] TIAN N, XU X, LI W. Multi-view comparative self-supervised graph learning for neuro-disease diagnosis using functional brain networks[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.

[71] DENG J, GUAN L, LI W. Topology-Enhanced Multi-Modal Residual Graph Convolutional Network for Autism Spectrum Disorder Identification[C]//2024 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2024: 1-6.

[72] MOU L, XIAO X, CAO W, et al. Efficient and accurate capsule networks with b-spline-based activation functions[C]//2024 International Conference on New Trends in Computational Intelligence. 2024.

[73] GUO J, LI W, HU J. Robust support vector machine based on sample screening[C]//The 14th International Conference on Information Science and Technology. 2024.

[74] ZHENG H, HU J, LI W. PageRank algorithm based on dynamic damping factor[C]//2023 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2023.

[75] LI T, CHEN X, LI W. An improved AdaBoost method in imbalanced data Learning[C]//2023 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2023.

[76] CHENG Z, CHEN X, LI H, et al. Minimum error entropy high-order extended Kalman filter[C]//2023 International Conference on Cyber-Physical Social Intelligence (ICCSI). 2023.

## Preprints and Other Works

[77] ZHENG Z, CHEN X, JIANG H, et al. Multi-Source Multi-View Graph Domain Adaptation with Hyperbolic Residual Encoding for Cross-Site MDD Identification from rs-fMRI[PP/OL]. arXiv (2026-07)[2026-10-05]. https://arxiv.org/abs/2607.29531.

[78] CUI L, WU Y, YING T, et al. Leveraging multi-source and heterogeneous signals for fatigue detection[PP/OL]. arXiv (2025-07)[2026-10-05]. https://arxiv.org/abs/2507.16859.

[79] LI W, CAO M, CHEN S. Jacobian norm for unsupervised source-free domain adaptation[PP/OL]. arXiv (2022-04)[2026-10-05]. https://arxiv.org/abs/2204.03467.

[80] LI W, GENG C, CHEN S. Leave zero out: Towards a no-cross-validation approach for model selection[PP/OL]. arXiv (2020-12)[2026-10-05]. https://arxiv.org/abs/2012.13309.

[81] YU J, WANG X, TU S, et al. Kola: Carefully benchmarking world knowledge of large language models[PP/OL]. arXiv (2023-06)[2026-10-05]. https://arxiv.org/abs/2306.09296.

[82] CHEN S, CHEN X, CAO W, et al. An Improved Efficient Algorithm for Spiking Neural Network Through Dataset Distillation[PP/OL]. SSRN[2026-10-05]. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4836784.

[83] XIAO X, CHEN X, LI W, et al. Graph Adversarial Discriminative Domain Adaptation with Category-Level Alignment[PP/OL]. SSRN[2026-10-05]. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4898772.

## Academic Service

1. Program Chair, CPSI 2027
2. Co-Program Chair, CPSI 2026
3. Area Chair, IEEE SMC 2026
4. Area Chair, IFAC CPHS 2020
5. Area Chair, ICCSI 2022–2025
6. Convener, Embodied Intelligence Conference Parallel Forum
7. Editorial Board Member, International Journal of Intelligent Control and Systems
8. Youth Editorial Board, Journal of Data Acquisition and Processing
9. Youth Editorial Board, Meta-Radiology
10. Youth Editorial Board, Command, Control and Simulation
11. Guest Editor, Frontiers in Neuroscience
12. Guest Editor, Frontiers in Aging Neuroscience
13. Guest Editor, Frontiers in Cell and Developmental Biology
14. Guest Editor, Frontiers in Pharmacology
15. Outstanding Reviewer, Journal of Data Acquisition and Processing
16. Youth Editorial Board, Aging & Disease
17. Youth Editorial Board, Brain-X
18. Youth Editorial Board, Cog
19. Youth Editorial Board, Artificial Intelligence Science and Engineering
20. Council Member, Shandong Association for Artificial Intelligence
21. Member, Young Scientists Professional Committee, Shandong Association for Artificial Intelligence
22. Member, Embodied Intelligence Professional Committee, Shandong Association for Artificial Intelligence
23. Corresponding Member, CAAI Machine Learning Committee
24. Deputy Secretary-General, CICC Embodied Intelligence Committee
25. Member, CAA Human–Machine Education Committee
26. Member, CSS Virtual Technology and Applications Committee

27. Committee Member, Youth Workshop on Medical Imaging
28. Council Member, Chongqing Mathematical Society

## Personal Recognition

1. 2026 / IEEE SMC 2026 Best Conference Paper Award Finalist
2. 2026 / CPSI Best Paper Award
3. 2025 / Taishan Scholars Young Expert
4. 2025 / CPSI Best Paper Award
5. 2025 / Outstanding Faculty Advisor, China Robot and AI Competition
6. 2025 / Third Prize, 5th Chongqing University Teachers' Teaching Innovation Competition (Intermediate and Below Rank Group)
7. 2023 / ICCSI Best Paper Finalist Award
8. 2022 / National Scholarship for Doctoral Students

## Student Awards

1. 2026 / Jingxuan Cui and Coauthors: CPSI 2026 Best Paper Award
2. 2025 / First Prize, China Robot and AI Competition National Finals
3. 2025 / First Prize, China Robot and AI Competition Chongqing Selection
4. 2025 / National Third Prize, National Digital Art Design Competition
5. 2025 / MCM/ICM Honorable Mention and Meritorious Winner
6. 2025 / First Prize, China International College Students' Innovation Competition: Paying Xianzhi
7. 2025 / First Prize, China International College Students' Innovation Competition: Yingluo Zhizhen
8. 2025 / Second Prize, iCAN Campus Competition
9. 2025 / Third Prize, China Graduate Electronics Design Competition, Southwest Region
10. 2024 / MCM/ICM Honorable Mention and Meritorious Winner
11. 2024 / First and Third Prizes, China International College Students' Innovation Competition, Campus Round
12. 2024 / Second Prize, China Robot and AI Competition National Finals
13. 2024 / Second Prize, Chongqing Robot Competition
14. 2024 / First Prize, National Business Elite Challenge Finals
15. 2024 / Third Prize, National Intelligent Transportation Innovation and Entrepreneurship Competition
16. 2024 / Third Prize, China Graduate Electronics Design Competition
17. 2024 / Second Prize, University Entrepreneurship Competition
18. 2024 / First Prize, RAICOM Robot Developer Competition
19. 2023 / First and Second Prizes, National Undergraduate Mathematical Contest in Modeling, Chongqing
20. 2023 / Third Prize, China Internet+ College Students' Innovation and Entrepreneurship Competition, Campus Round
21. 2023 / Third Prize, China Software Cup Western Division